

MTH601 QUIZ(3)

Module: 7,8,9,10

RIZ MUGHAL

SQA ENGINEER:

I'm providing 100% correct quiz solution.

You can visit my YouTube channel for more quiz solution, also final year project including project assignments, and viva.

Hira Mumtaz

YOUTUBE:

https://www.youtube.com/channel/UCINsFwDiB62SValCcPDZbRQ/playlists

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MTH601:Quiz

Quiz Start Time: 02:17 PM, 12 Februar

Question # 1 of 10 (Start time: 02:17:53 PM, 12 February 2021) **Total M**

In the Least Cost Method, for any cell both demand and supply are satisfied then which of the following will be crossed out?

Select the correct option

- ○ Only row
- ○ Only column
- ○ Both row and column
- ◉ Either of row or column

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MTH601:Quiz

Quiz Start Time: 02:17 PM, 12 February 20

Question # 2 of 10 (Start time: 02:18:13 PM, 12 February 2021) **Total Marks**

For North West Corner method, in the first row and first column, each source and sink contain '5' units ; then after allocating the appropriate amount 'x11' in the cell (1,1), we will move towards which of the following cell?

Select the correct option

- (●) Arbitrarily either towards (1,2) or (2,1)
- ( ) Towards (1,2) if 1st row contains more occupied cells than 2nd row
- ( ) Towards (2,1) if 2nd row contains more occupied cells than 1st row
- ( ) Neither towards (1,2) nor (2,1) but move diagonally towards (2,2)

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Question # 3 of 10 (Start time: 02:18:27 PM, 12 February 2021)

The Transportation Problem is one of the sub-class of ----------.

Select the correct option

- (●) Linear Programming Problem
- ( ) Inventory Control Problem
- ( ) Network Flow Problem
- ( ) Stochastic Problem

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Quiz Start Time: 02:17 PM, 12 February 20

Question # 4 of 10 (Start time: 02:18:41 PM, 12 February 2021) **Total Marks**

For an assignment problem, if the numbers of agents and tasks are _____, then the problem is called _____ assignment.

Select the correct option

- (●) equal, balanced
- ( ) unequal, balanced
- ( ) equal, unbalanced
- ( ) none of these

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Quiz Start Time: 0

Question # 5 of 10 (Start time: 02:18:55 PM, 12 February 2021)

The North - West Corner Rule

Select the correct option

- ○ Is based on the concept of minimizing opportunity cost.
- ○ Is used to find optimal solution
- ◉ Is used to find an initial feasible solution
- ○ None of above

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MTH601:Quiz

Quiz Start Time: 02:17 PM, 12 February 202

Question # 6 of 10 (Start time: 02:19:08 PM, 12 February 2021) **Total Marks:**

Which of the following will be the penalty cost of a row in Vogel's approximation method, if the smallest element is '3' and other elements are 3, 4, 5 ?

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Select the correct option

- (●) 0
- ( ) 1
- ( ) 2
- ( ) 3

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Question # 7 of 10 (Start time: 02:19:22 PM, 12 February 2021)

In a Transportation Problem, the objective function 'Z' gives ----------.

Select the correct option

- (●) Total Cost of transportation
- ( ) Total Time of transportation
- ( ) Total Profit of transportation
- ( ) Total inventory be supplied in transportation

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Question # 8 of 10 (Start time: 02:19:36 PM, 12 February 2021)

In the assignment problem, the cost matrix is ----------.

Select the correct option

- rectangular
- square
- non-singular
- singular

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Quiz Start Time: 02:17 PM, 12 Febru

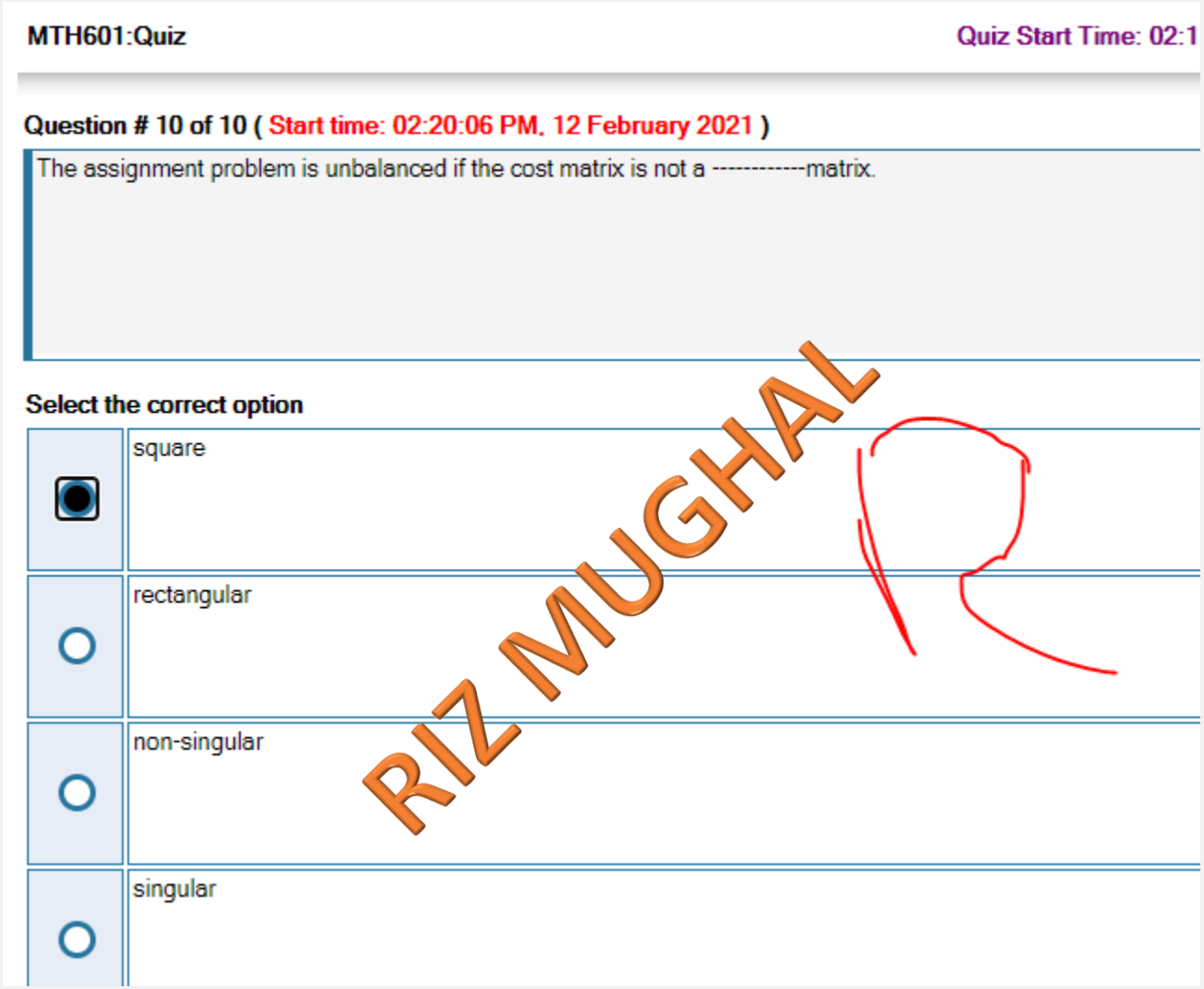
Question # 9 of 10 (Start time: 02:19:50 PM, 12 February 2021) **Total**

In which of the following method to solve a Transportation problem, we have to calculate one more row and one more column?

Select the correct option

- ○ North West Corner method
- ○ Least Cost Method
- ◉ Vogel's approximation Metho
- ○ Simplex Method

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Quiz Start Time: 02:1
Question # 10 of 10 (Start time: 02:20:06 PM, 12 February 2021)
The assignment problem is unbalanced if the cost matrix is not a ----------matrix.
Select the correct option
square
rectangular
non-singular
singular
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2nd account

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Quiz Start Time: 02:43 PM, 12 February 20

Question # 1 of 10 (Start time: 02:43:28 PM, 12 February 2021) **Total Marks**

If the cost matrix in an Assignment problem is not square then which the following modification will be made to balance the given problem?

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Select the correct option

- Add a dummy row(column) with positive cost elements
- Add a dummy row(column) with negative cost elements
- Add a dummy row(column) with zero cost elements
- Add a dummy row(column) with arbitrary(including negative or positive) cost elements

MTH601:Quiz Quiz Start Time: (

Question # 2 of 10 (Start time: 02:43:47 PM, 12 February 2021)

The North - West Corner Rule

Select the correct option

- ○ Is based on the concept of minimizing opportunity cost.
- ○ Is used to find optimal solution
- ◉ Is used to find an initial feasible solution
- ○ None of above

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Quiz Start Time: 02:43 PM, 12 Febr

Question # 3 of 10 (Start time: 02:44:01 PM, 12 February 2021) Tota

If a balanced Transportation problem with 'm' sources and 'n' sinks then degeneracy will arise only if there are less than ---------- independent allocations in the solution.

Select the correct option

- m-n-3
- m-n-2
- m-n-1
- m-n-4

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Quiz Start Time: 02:43 PM, 12 Fe

Question # 4 of 10 (Start time: 02:44:21 PM, 12 February 2021)

The column, which is introduced in the matrix to balance an unbalanced Transportation problem, is known as-------

Select the correct option

- key column
- idle column
- slack column
- dummy Column

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Quiz Start Time: 02:43 PM, 12 February

Question # 5 of 10 (Start time: 02:44:39 PM, 12 February 2021) **Total Ma**

The solution of a transportation problem with m rows (supplies) and n (destinations) is feasible if numbers of positive allocations are

Select the correct option

- ○ m+n
- ◉ m+n-1
- ○ m-n+1
- ○ m+n+1

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Quiz Start Time: 02:43 PM, 12 February 2

Question # 6 of 10 (Start time: 02:44:53 PM, 12 February 2021) **Total Mar**

For an unbalanced Transportation problem, if the total demand is LESS than total supply then which of the following is true in order to balance the problem?

Select the correct option

- (●) A dummy sink would have to include with demand equal to the s[illegible]
- ( ) A dummy source would have to include with supply equal to shortage
- ( ) One constraint will have evacuate
- ( ) One constraint will have to add

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Quiz Start Time: 02

Question # 7 of 10 (Start time: 02:45:08 PM, 12 February 2021)

There always exists a non-degenerate Feasible solution of a Balanced Transportation Problem.

Select the correct option

- ○ True
- ◉ False

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Question # 8 of 10 (Start time: 02:45:25 PM, 12 February 2021)

To solve the degeneracy in a transportation problem we have to----------.

Select the correct option

- ○ put allocation in one of the empty cell with zero valu
- ○ put a small element epsilon in any one of the empty cell
- ◉ allocate the smallest element epsilon in such a cell
- ○ apply duality principle

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MTH601:Quiz Quiz Start Time: 02:43 PM, 12 February 2021

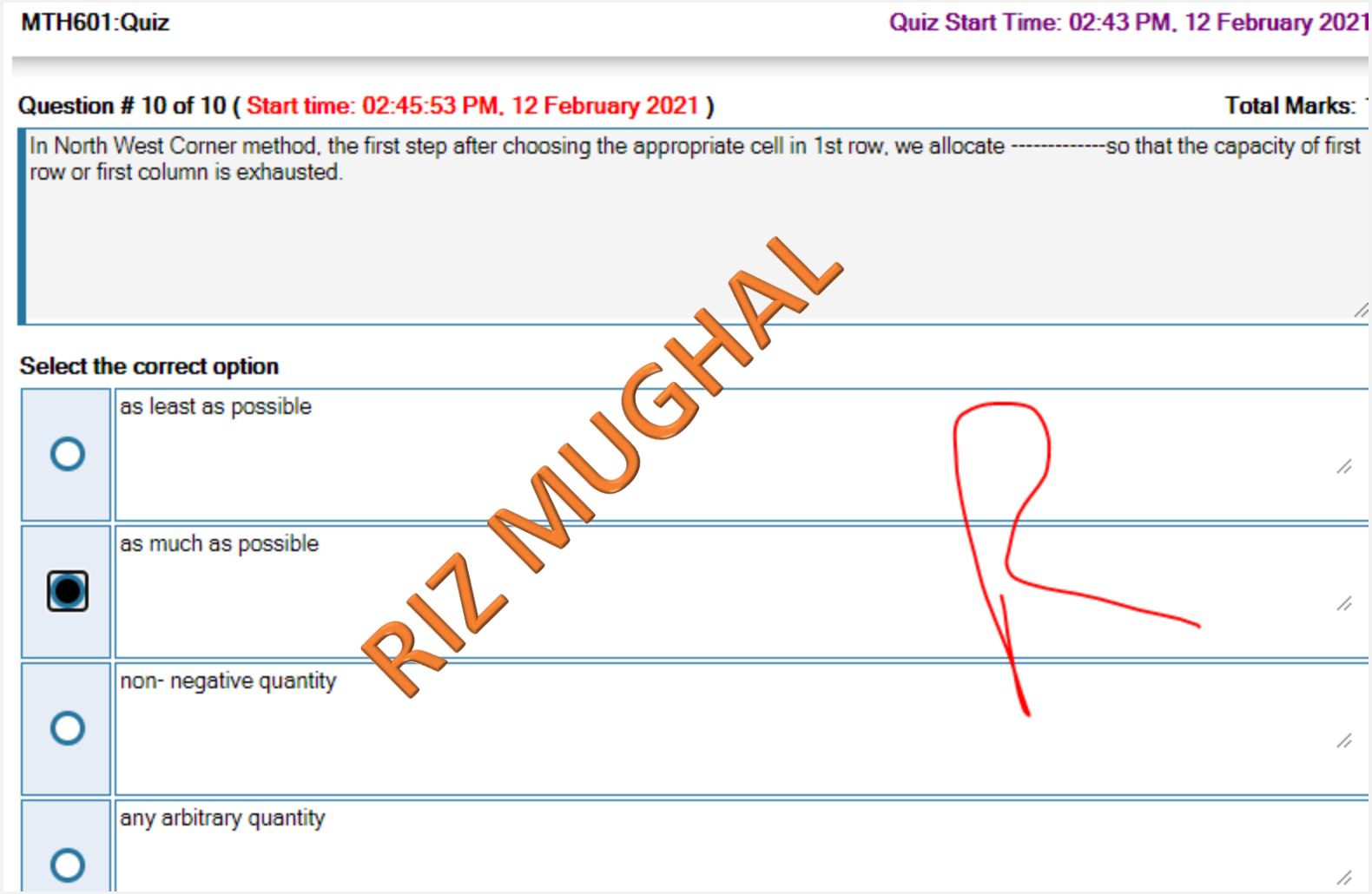
Question # 9 of 10 (Start time: 02:45:39 PM, 12 February 2021) **Total Marks:** 1

In the Vogel's approximation Method for solving a Transportation problem, Penalty measure for any row or column, is given by which of the following?

Select the correct option

- Difference between the largest unit cost to the next largest cost in the same row(column)
- Difference between the smallest unit cost to the next smallest cost in the same row(column)
- Difference between the Largest unit cost to the smallest unit cost in the came row(column)
- Difference between any two arbitrary costs in the came row(column)

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MTH601:Quiz
Quiz Start Time: 02:43 PM, 12 February 2021
Question # 10 of 10 (Start time: 02:45:53 PM, 12 February 2021)
Total Marks:
In North West Corner method, the first step after choosing the appropriate cell in 1st row, we allocate ----------so that the capacity of first row or first column is exhausted.
Select the correct option
as least as possible
as much as possible
non- negative quantity
any arbitrary quantity
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3rd account

MTH601:Quiz Quiz Start Time: 03:51 PM, 12 February 2021

Question # 1 of 10 (Start time: 03:51:44 PM, 12 February 2021) **Total Marks:**

Under which of the following condition in the Least Cost Method, after first allocation in cell (1,1), next allocation along the diagonal in cell (2,2) can be made?

Select the correct option

- ○ Total Supply = Total Demand
- ○ If in the second row and second column, each resource and sink contain equal units
- ● If after crossed out 1st row & 1st column, cell (2,2) is of minimum cost
- ○ If Total Supply is not equal to Total Demand

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MTH601:Quiz

Quiz Start Time: 03:51 PM, 12 February 2

Question # 2 of 10 (Start time: 03:52:05 PM, 12 February 2021)

Total Mar

The solution of a transportation problem with m rows (supplies) and n (destinations) is feasible if numbers of positive allocations are

Select the correct option

- ○ m+n
- ◉ m+n-1
- ○ m-n+1
- ○ m+n+1

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MTH601:Quiz

Quiz Start Time: 03:51 PM, 12 Febru

Question # 3 of 10 (Start time: 03:52:32 PM, 12 February 2021) **Total**

Due to which of the following reason, Simplex method is not preferred to solve a Transportation problem?

Select the correct option

- (●) Since it contains large number of decision variables 'xij's' so that it becomes complicated
- ( ) Since Transportation problem contains constraints of '=' type
- ( ) Since Transportation problem does not contain positive slacks
- ( ) Since Transportation problem does not contain negative slacks

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MTH601:Quiz

Quiz Start Time: 03:51 PM, 12 February 2021

Question # 4 of 10 (Start time: 03:52:45 PM, 12 February 2021) **Total Marks:** 1

Which values of "a_{ij}'s" in the standard form of a linear programming problem will transform it in standard Transportation problem?

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Select the correct option

Reload Math Equations

- ○ $a_{ij} = 0$ for all $i = 1, 2, \cdots, m; j = 1, 2, \cdots, n$
- ○ $a_{ij} = 1$ for all $i = 1, 2, \cdots, m; j = 1, 2, \cdots, n$
- ○ $a_{ij} = 0$ and 1 for all $i = 1, 2, \cdots, m; j = 1, 2, \cdots, n$
- ● $a_{ij} = 0$ or 1 for all $i = 1, 2, \cdots, m; j = 1, 2, \cdots, n$

MTH601:Quiz Quiz Start Time: 03:51 PM, 12 Fe

Question # 5 of 10 (Start time: 03:53:02 PM, 12 February 2021) T

In which of the following method to solve a Transportation problem, allocation in a cell depends upon penalty cost?

Select the correct option

- ○ North West Corner Method
- ◉ Vogel's approximation Method
- ○ Least Cost Method
- ○ Simplex Method

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Quiz Start Time:

Question # 6 of 10 (Start time: 03:53:16 PM, 12 February 2021)

If a Transportation problem has 'm+n' constraints then how many number of unknowns it will have?

Select the correct option

- [x] m.n
- [ ] m+n
- [ ] m-n
- [ ] m/n

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Question # 7 of 10 (Start time: 03:53:28 PM, 12 February 2021)

The North - West Corner Rule

Select the correct option

- ○ Is based on the concept of minimizing opportunity cost.
- ○ Is used to find optimal solution
- ◉ Is used to find an initial feasible solution
- ○ None of above

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Quiz Start Time

Question # 8 of 10 (Start time: 03:53:47 PM, 12 February 2021)

The assignment problem is unbalanced if the cost matrix is not a ----------matrix.

Select the correct option

- (●) square
- ( ) rectangular
- ( ) non-singular
- ( ) singular

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Question # 9 of 10 (Start time: 03:54:06 PM, 12 February 2021)

Transportation problem is basically a --------

Select the correct option

- Maximization model
- Minimization model
- Transshipment problem
- Iconic model

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Quiz Start Time: 03:51 PM, 12

Question # 10 of 10 (Start time: 03:54:21 PM, 12 February 2021)

The column, which is introduced in the matrix to balance an unbalanced Transportation problem, is known as-------

Select the correct option

- ○ key column
- ○ idle column
- ○ slack column
- ◉ dummy Column

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Thank you for watching ☺

Share with your fellows

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